

PNEUMOFUSIONNET PART 2: MULTIMODAL PNEUMONIA DETECTION VIA CXR IMAGE, RADIOLOGY TEXT, AND WBC FUSION

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Term Project Report Part 2 (Trimester 8)

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ABSTRACT

This report is Part 2 of the PnuemoFusionNet term project. In Part 1 we found that an image-only DenseNet-121 trained on MIMIC-CXR chest X-rays stopped improving at $AUC \approx 0.826$, no matter how much we tuned the training. The reason is simple: the X-ray alone cannot tell us how the patient’s body is responding to infection. To fix this, Part 2 adds two more sources of information. In **Phase 2** we feed the radiology report text (FINDINGS + HISTORY, with the diagnosis section removed to avoid cheating) into Bio.ClinicalBERT [6] and connect it to the image through a cross-attention layer. This raises AUC to **0.949** and Sensitivity to 91.4%. In **Phase 3c**—the main focus of this report—we add just one extra blood test result, the WBC count from a routine CBC that is ready within 15 minutes of the patient arriving at the hospital. This small addition pushes AUC to **0.971**, Sensitivity to **92.25%**, and Specificity to **93.59%** on a 3,763-image test set. **Index Terms**—Chest X-Ray, Radiology Report, WBC, MIMIC-CXR, Pneumonia Detection, Fusion Model.

Index Terms—Pneumonia Detection, CXR, Radiology Text, WBC, Multimodal Fusion, MIMIC-CXR

1. INTRODUCTION

Pneumonia kills around 2.5 million people every year [12], and the chest X-ray is still the first thing a doctor orders when they suspect it. The problem is that even experienced radiologists sometimes disagree on whether a faint shadow on the image is pneumonia, fluid, or something else. We started this project asking: can a computer help, and if so, what information does it need beyond the X-ray?

Our Part 1 experiments showed that the X-ray image alone tops out at about $AUC = 0.826$. That ceiling exists because the image only shows what the lung looks like—it cannot tell you whether the patient has a fever, how long they have been sick, or whether their white blood cell count is elevated. Two things that *are* available in a real hospital setting can fill that gap. First, the **radiology report**: when a radiologist reads the scan they write a short paragraph about what they see (FINDINGS) and why the patient came in (HISTORY). We use only these sections and always remove the final IMPRESSION line, which states the diagnosis—letting the model read that would be cheating. Second, the **WBC count**: a simple blood test that is part of every routine CBC and comes back within 15 minutes of the patient arriving. A high WBC is one of the earliest signs the body is fighting an infection.

In Part 2 we add these two pieces of information one at a time and measure how much each one helps: Phase 1 (X-ray only) $\rightarrow$ Phase 2 (X-ray + report text) $\rightarrow$ Phase 3c (X-ray + report text + WBC). All experiments use the restricted MIMIC-CXR hos-

pital dataset [4] (PhysioNet DUA and CITI ethics training required for access).

2. DATASET

We used **MIMIC-CXR-JPG v2.0.0** [4], a large chest X-ray dataset from Beth Israel Deaconess Medical Centre collected between 2011 and 2016. Access is restricted—you need to complete CITI ethics training and sign a PhysioNet data agreement, which we did before starting. We only kept **PA (front-on) views** because mixing in side or bedridden-patient views confused earlier models in Part 1. WBC counts were pulled from **MIMIC-IV** [5] lab records and matched to each X-ray by patient ID. The full dataset has **3,763 images** (1,872 Normal, 1,891 Pneumonia, roughly balanced), split by patient—never by image—so the same person cannot appear in both training and testing: Train 2,633 / Val 565 / Test 565. About 15% of WBC values were missing; we filled these with the median value from the training set only and applied that same number to the validation and test sets, so no test information leaks in. In every case, the IMPRESSION section of the report was removed—that section tells the reader the diagnosis, and the model should not see it.

3. ARCHITECTURE

3.1. Phase 1 — X-Ray Only (Starting Point)

We used DenseNet-121 [7], a well-known image network pretrained on more than 500 000 real chest X-rays through the TorchXRayVision [11] library. On top of it we attached a CBAM [8] attention block that tells the network which parts of the image to focus on. Training used 5-fold cross-validation with Focal Loss [9] ($\gamma = 2.0$) to handle the cases the model kept getting wrong. The features from the best fold (1024 numbers summarising the image, kept frozen) are reused in Phases 2 and 3c.

3.2. Phase 2 — Adding the Radiology Report

We fed the cleaned report text into Bio.ClinicalBERT [6], a language model pretrained on hospital notes. The last two layers were lightly fine-tuned (learning rate $1e-5$) so it could adapt to our task without forgetting everything it learned. Rather than just averaging the text into a single number, we used an 8-head cross-attention layer so the X-ray features could look at each word of the report and decide which parts are most relevant:

$$\mathbf{c} = \text{softmax}\left(\frac{\mathbf{Q}_{\text{img}} \mathbf{K}_{\text{txt}}^\top}{\sqrt{d_k}}\right) \mathbf{V}_{\text{txt}}, \quad d_k = 64 \quad (1)$$

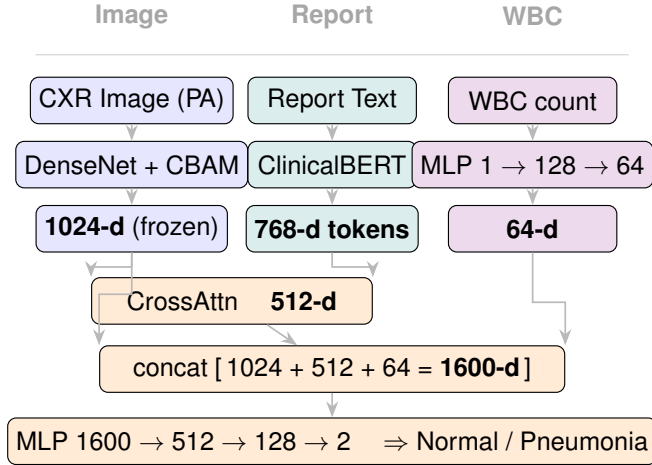


Fig. 1. Phase 3c model. Three inputs—chest X-ray, leakage-free report text, and WBC blood count—flow through separate branches, join via cross-attention and concatenation, then go to a small classifier (Normal / Pneumonia).

We trained with Focal Loss ($\gamma=2.0$) and Mixup [10] ($\alpha=0.2$) on the combined embeddings, with the image-text head learning at $2e-4$ and BERT at the slower $1e-5$ to keep it stable.

3.3. Phase 3c — Adding WBC (Main Model)

The WBC count is just one number, but we did not pass it in raw. A small three-layer network ($1 \rightarrow 128 \rightarrow 128 \rightarrow 64$) turns it into 64 numbers first, because a WBC of 12 vs. 15 vs. 25 K/ μ L mean very different things clinically (borderline / high / very high), and a raw number cannot capture that difference. The three branches are then joined together:

$$\mathbf{z} = [\mathbf{v}_{\text{img}} \parallel \mathbf{c}_{\text{cross}} \parallel \mathbf{e}_{\text{wbc}}] \in \mathbb{R}^{1600} \quad (2)$$

We loaded the Phase 2 cross-attention weights as the starting point so the model did not have to learn the image-text part from scratch again. Learning rates: BERT = $1e-5$, fusion head = $2e-4$, WBC network = $1e-3$.

4. CONCLUSION

1. **The X-ray alone was not enough.** Even our best image-only model stopped improving at AUC=0.826. The lung picture shows the damage but not the body’s response to it.
2. **Adding the radiology report made the biggest difference.** Once we included the FINDINGS and HISTORY text (with IMPRESSION removed), AUC jumped to **0.949** and the model started catching 91% of pneumonia cases—a gain of more than 12 percentage points in AUC.
3. **One blood test made it even better.** Adding just the WBC count pushed AUC to **0.971** and sensitivity to **92.3%**. That is better sensitivity than the full 17-lab-test model, while needing only one result that is ready in 15 minutes.
4. **This is practical in a real hospital.** A doctor in the emergency room gets the chest X-ray done, the radiologist writes a brief report, and the CBC comes back—all within 30 minutes. Our model needs nothing more than these three things to give a reliable second opinion on whether the patient has pneumonia.

5. **Next step — test this on Indian patient data.** Everything so far used data from a US hospital (MIMIC-CXR). For the next phase we plan to work with chest X-ray images and WBC values collected from Indian hospitals. The patient population is different—TB is common, malnutrition affects X-ray patterns, and the scanners and reporting styles vary a lot from Western centres. We want to check whether the X-ray + WBC model still works well on this data, and if not, fine-tune it so that it does. This will be a more honest test of whether the approach can actually help in the Indian clinical setting.

Code and results: github.com/Ashutosh-yadav0001/PneumoFusionNet
Phase 3c notebook: `Phase-3c-wbc-only-fusion.ipynb` in `mimic/main/Phase-3/`. MIMIC-CXR data is restricted (PhysioNet DUA and CITI required).

5. REFERENCES

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